

FitCore

Gym Management System Documentation & Marketing Plan

Table of Contents

1. Project Overview
2. Project Objectives
3. Target Users
4. Scope
5. System Features
6. Non-Functional Requirements
7. System Architecture
8. Database Schema
9. API Endpoints
10. User Interface (UI/UX)
11. Payments & Attendance Integration
12. Security & Compliance
13. Development Roadmap
14. Testing & Quality Assurance
15. Marketing Plan
16. Branding & Identity
17. Pricing & Distribution
18. Key Performance Indicators (KPIs)
19. Initial Data
20. Risks & Mitigation
21. Conclusion

1. Project Overview

FitCore is a comprehensive Gym Management Web Application designed to streamline and automate the daily operations of modern gyms and fitness centers. The platform integrates every essential business function—including membership management, class scheduling, private training sessions, attendance tracking, retail shop management, inventory control, online payments, reporting, and notifications—within a single role-based system.

Built using **.NET 9 (C#)**, **SQL Server**, and **Entity Framework Core**, FitCore provides secure, scalable, and maintainable solutions for gym owners, administrators, trainers, receptionists, and members.

2. Project Objectives

FitCore aims to:

- Replace manual and spreadsheet-based gym operations with a centralized digital platform.
 - Enable members to independently manage bookings, payments, attendance, and shopping.
 - Provide administrators with complete operational visibility through dashboards and analytics.
 - Automate repetitive business tasks such as membership expiry notifications and low-stock alerts.
 - Improve business efficiency through secure, role-based access and complete audit logging.
 - Support future scalability for multi-branch gyms and AI-powered features.
-

3. Target Users

The system is designed for:

Gym Owners & Administrators

Managing memberships, finances, reports, staff, inventory, and overall business operations.

Receptionists

Managing front-desk activities including member registration, check-in, payments, and customer support.

Trainers

Managing classes, private sessions, schedules, and working hours.

Members

Booking memberships, classes, private sessions, shopping products, making payments, and checking attendance.

4. Project Scope

In Scope (MVP)

- Authentication & Authorization
- Membership Management
- Class Scheduling & Booking
- Private Training Sessions
- Gym Services & Packages
- Attendance (QR & Manual Check-In)
- Shop & Inventory Management
- Payment Processing
- Invoice Generation
- Notifications
- Audit Logging
- Admin Dashboard & Reports
- Trainer Working Hours

Out of Scope (Future Versions)

- Native Mobile Applications
 - Multi-Branch Management
 - Wearable Device Integration
 - AI-Based Workout Recommendations
 - Loyalty & Rewards Programs
-

5. System Features

Authentication & User Management

- JWT Authentication
 - Password Hashing
 - Role-Based Authorization
 - Role Change Request Workflow
-

Membership Management

- Membership Plans
 - Subscription Activation
 - Freeze / Unfreeze
 - Expiry Management
 - Membership History
-

Class Management

- Browse Classes
- Book Classes
- Cancel Booking
- Capacity Control
- Booking History

Private Training

- Session Requests
- Trainer Assignment
- Cancellation
- Completion Tracking

Attendance Management

- QR Code Check-In
- Manual Check-In
- Attendance History
- Attendance Statistics
- Duplicate Check Prevention

Shop & Inventory

- Product Catalog
- Categories
- Shopping Cart
- Checkout
- Inventory Tracking
- Suppliers
- Stock Transactions

Payments

Supported payment methods:

- Stripe Checkout

- Cash
- InstaPay
- PayPal

Features include:

- Payment Recording
 - Invoice Generation
 - Stripe Webhooks
 - Payment History
-

Notifications

Automatic notifications for:

- Membership Expiry
 - Low Stock
 - New Announcements
 - Booking Reminders
-

Dashboard

Provides analytics including:

- Revenue
 - Membership Distribution
 - Attendance
 - Product Sales
 - Recent Notifications
 - Exportable Reports
-

Audit Logs

Tracks:

- User
- Action
- Entity
- Timestamp

Supports advanced filtering.

6. Non-Functional Requirements

Performance

Fast API response times under concurrent usage.

Scalability

Layered architecture with support for background processing.

Security

- HTTPS
- JWT Authentication
- Password Hashing
- Role-Based Authorization

Maintainability

- DTO Pattern
- Repository Pattern
- Service Layer
- Unit-Testable Business Logic

Auditability

All sensitive operations are logged.

7. System Architecture

Frontend

Role-based dashboards served from the API project using:

- HTML
- CSS
- JavaScript

Separate layouts are provided for:

- Admin
 - Receptionist
 - Trainer
 - Member
-

Backend

Built using:

- .NET 9 Web API
- C#
- ASP.NET Core

Project Layers:

- FitCore.API
 - FitCore.BLL
 - FitCore.DAL
 - FitCore.Shared
-

Database

- SQL Server
- Entity Framework Core
- Code-First Migrations

Authentication

- JWT Tokens
- Custom Identity
- Role Claims

Background Jobs

Hangfire handles:

- Membership Expiry
- Notification Scheduling
- Automated Tasks

Storage

- Local wwwroot
- Ready for Azure Blob Storage or Amazon S3

8. Main Database Tables

- Users
- UserRole
- MemberProfile
- Trainer
- TrainerWorkingHour
- Membership
- GymService
- Class
- ClassSchedule

- Booking
- PrivateSession
- Attendance
- Product
- Category
- Cart
- CartItems
- Inventory
- InventoryTransaction
- InventoryTransactionItem
- Supplier
- Invoice
- InvoiceItem
- Payment
- Notification
- AuditLog
- RoleChangeRequest

9. Sample API Endpoints

Authentication

- POST /api/auth/login
- POST /api/auth/register-member
- POST /api/auth/create-staff

Classes

- GET /api/classes/browse
- POST /api/classes/book

Private Sessions

- POST /api/privatesessions

Attendance

- GET /api/attendance/me/qrcode
- POST /api/attendance/checkin/scan

Shop

- GET /api/shop/products
- POST /api/shop/checkout

Payments

- POST /api/payment/create-checkout-session

Dashboard

- GET /api/admin/dashboard/stats

Audit Logs

- GET /api/auditlogs

10. User Interface (UI/UX)

The application includes:

- Landing Page
- Login & Registration
- Admin Dashboard
- Reception Dashboard
- Trainer Dashboard
- Member Dashboard
- QR Digital Pass
- Shop & Checkout Pages
- Success & Failed Payment Pages

Each role has a customized navigation sidebar and header.

11. Payments & Attendance Integration

Payments

Stripe Checkout Sessions process online payments securely.

Alternative payment methods include:

- Cash
- InstaPay
- PayPal

Attendance

Members receive a personal QR Code used for contactless check-in.

Hangfire automates:

- Membership Expiry Checks
 - Notification Delivery
-

12. Security & Compliance

FitCore follows modern security practices:

- HTTPS Everywhere
 - JWT Authentication
 - Password Hashing
 - Role Authorization
 - Audit Logging
 - Stripe PCI Compliance
-

13. Development Roadmap

Phase 1

Preparation (2 Weeks)

- Requirements Analysis
- Database Design
- Architecture Setup

Core Development (8 Weeks)

Week 1–2

- Authentication
- Roles
- Database

Week 3–4

- Memberships
- Classes
- Bookings

Week 5–6

- Attendance
- Shop
- Inventory
- Payments

Week 7

- Dashboard
- Notifications
- Audit Logs

Week 8

- Testing
- Deployment
- UAT

Phase 2

- Multi-Branch Support
 - Mobile Applications
 - AI Features
 - Loyalty System
-

14. Testing & Quality Assurance

Testing includes:

- Unit Testing
 - Integration Testing
 - Manual Role-Based Testing
 - Stripe Sandbox Testing
 - Hangfire Job Verification
-

15. Marketing Plan

Target Market

FitCore primarily targets:

- Independent Gyms
- Boutique Fitness Studios
- Multi-Service Fitness Centers
- Growing Gym Chains

The platform is especially suitable for organizations still relying on spreadsheets or disconnected software solutions.

Customer Needs

Customers typically seek:

- Faster membership management

- Efficient attendance tracking
 - Better trainer scheduling
 - Accurate inventory control
 - Revenue insights
 - Local payment flexibility
-

Competitor Analysis

Main competitors include international gym management platforms, generic scheduling software, and manual operational methods.

FitCore differentiates itself by combining:

- Memberships
- Bookings
- Attendance
- Inventory
- Retail Shop
- Payments

within one unified platform while supporting local payment methods such as Cash and InstaPay alongside Stripe.

Marketing Strategy

Short-Term Goals (First 6 Months)

- Onboard 15–20 pilot gyms.
- Publish three customer case studies.
- Build partnerships with gym equipment suppliers.

Long-Term Goals (12–18 Months)

- Become a leading regional gym management platform.
- Expand localization.

- Introduce AI-powered analytics.
 - Reach more than 100 active gym clients.
-

Promotion Channels

Marketing activities include:

- LinkedIn
- Facebook
- Instagram
- SEO Blog Content
- YouTube Product Demonstrations
- Fitness Industry Exhibitions
- Referral Programs
- Live Webinars

Marketing Message

"Run your gym smarter — memberships, classes, and payments in one place."

16. Branding & Identity

Logo Concept

A dumbbell integrated with an upward heartbeat line.

Color Palette

- Red
- Black
- White

Brand Personality

- Professional
- Motivational

- Reliable

Core Message

"One platform. Every rep of your gym business."

17. Pricing & Distribution

Pricing

- Free Trial
- Starter Plan
- Growth Plan
- Pro Plan

Pricing scales based on member count and branch count.

Distribution

- Direct B2B Sales
 - Online Self-Service Registration
 - Equipment Supplier Partnerships
 - Future Referral Network
-

18. Key Performance Indicators (KPIs)

Business performance is measured through:

- Active Memberships
- Monthly Recurring Revenue (MRR)
- Trial-to-Paid Conversion Rate
- Class Booking Rate
- Private Session Booking Rate
- Attendance Frequency
- Shop Sales Revenue

- Customer Retention
 - Churn Rate
 - Trainer Utilization
 - Feature Adoption Rate
 - Net Promoter Score (NPS)
-

19. Initial Data

The system includes seed data for:

- Default User Roles
- Initial Administrator Account
- Membership Plans
- Sample Classes
- Products

Additional operational data is entered through the Admin Dashboard.

20. Risks & Mitigation

Risk	Mitigation
Payment Gateway Downtime	Allow Cash/InstaPay recording until service resumes
Inventory Inconsistencies	Transactional stock movements with audit logging
Scalability Challenges	Layered architecture supporting future caching and scaling
Scope Creep	Strict MVP boundaries with Phase 2 backlog
Security Threats	HTTPS, secure secret management, JWT authentication, and continuous audit logging

21. Conclusion

FitCore is a complete, scalable, and secure Gym Management System that combines operational management with business intelligence in one unified platform. By integrating memberships, scheduling, attendance, payments, inventory, reporting, and customer engagement into a centralized solution, FitCore enables fitness businesses to improve operational efficiency, enhance member experience, and support long-term growth. The platform is also designed with extensibility in mind, allowing future expansion into multi-branch management, mobile applications, and AI-powered fitness services without requiring major architectural changes.